

Abstract No.

Adding a Metabolic Layer to Glioblastoma Pathology with Label-Free NADH-Fluorescence Lifetime Imaging

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Background: Glioblastoma (GBM) is characterized by profound spatial heterogeneity that cannot be fully captured by conventional histopathology. While hematoxylin-eosin (H&E) staining and immunohistochemistry define tissue architecture and molecular markers, they fail to reveal the metabolic states underlying tumor progression and therapeutic response. We investigated whether label-free NADH Fluorescence Lifetime Imaging Microscopy (NADH-FLIM) could provide an additional functional layer of information by mapping metabolic heterogeneity directly in clinical specimens.

Methods: NADH-FLIM was applied to archived formalin-fixed paraffin-embedded (FFPE) GBM sections. Phasor analysis of endogenous NADH autofluorescence was used to quantify the free/bound NADH fraction (fB-NADH), enabling spatial characterization of oxidative phosphorylation (OXPHOS) and glycolytic metabolism across distinct histopathological regions. Metabolic profiles were further correlated with genetic drivers by immunohistochemistry.

Results: NADH-FLIM identified distinct, statistically significant metabolic signatures associated with major GBM hallmarks. Regions of microvascular proliferation showed significantly higher fB-NADH than adjacent normal-appearing tissue ($p < 0.0001$), consistent with an OXPHOS-dominant phenotype. Pseudopalisading cells surrounding necrosis displayed a significantly lower fB-NADH than the tumor core ($p < 0.001$), indicating a glycolytic shift associated with hypoxic adaptation. Isolated infiltrating tumor cells in perineural and perivascular satellitosis, as well as within histologically normal-appearing brain parenchyma, showed a shift toward a more glycolytic fB-NADH profile compared with the tumor core ($p < 0.001$). FGFR3-positive tumor cells showed a shift toward OXPHOS metabolism compared with adjacent FGFR3-negative cells.

Conclusions: NADH-FLIM complements conventional pathology by providing a quantitative, label-free functional map of GBM metabolism while preserving tissue architecture. This approach uncovers metabolically distinct niches that are invisible to routine histology and links oncogenic alterations, including FGFR3::TACC3 fusion, to bioenergetic states. These findings support NADH-FLIM as a promising platform for precision neuropathology and future AI-assisted diagnostic and predictive workflows.



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